```
import numpy as np
import matplotlib.pyplot as plt

# Experimental data
m = np.array([100, 200, 300, 400, 500])  # Suspended mass (g)

I = np.array([0.0297, 0.0377, 0.0303, 0.0355, 0.0315])  # Moment of inertia (kg m²)

# Mean measured moment of inertia
mean_I = np.mean(I)

# Theoretical moment of inertia
M = 23  # Assumed mass in kg
R = 0.1  # Radius in metres (assumed)

theoretical_I = (1/2) * M * R**2

# Create plot
plt.figure(figsize=(10, 4))

# Theoretical value (blue dashed line)
plt.axhline(
    y=theoretical_I,
    color='tab:blue',
    linestyle='--',
    linewidth=1.8,
    label=r'$\frac{1}{2}MR^2$ (M = 23 kg, assumed)'
)

# Mean measured value (red solid line)
plt.axhline(
    y=mean_I,
    color='red',
    linewidth=2,
    label='Mean measured I'
)

# Experimental data points
plt.scatter(
    m,
    I,
    color='black',
    s=50,
    label='Measured I (each load)',
    zorder=3
)

# Value labels above each point
for x, y in zip(m, I):
    plt.text(
        x, y + 0.004,
        f'{y:.4f}',
        ha='center',
        fontsize=9
    )

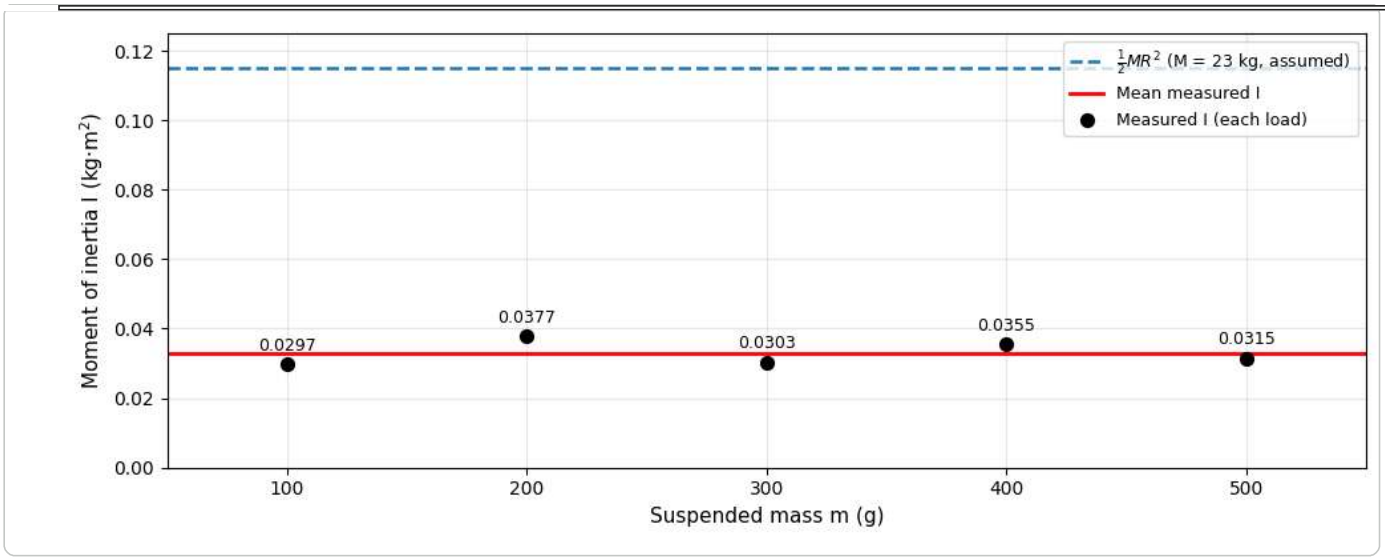
# Axis labels
plt.xlabel('Suspended mass m (g)', fontsize=11)
plt.ylabel(r'Moment of inertia I (kg·m$^2$)', fontsize=11)

# Axis limits
plt.xlim(50, 550)
plt.ylim(0, 0.125)

# Grid
plt.grid(True, alpha=0.3)

# Legend
plt.legend(loc='upper right', fontsize=9)

# Layout and display
plt.tight_layout()
plt.show()
```


$\frac{1}{2}MR^2$ (M = 23 kg, assumed)
Mean measured I
Measured I (each load)
Moment of inertia I (kg·m²)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0.0297
0.0377
0.0303
0.0355
0.0315
100
200
300
400
500
Suspended mass m (g)